

ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA
CAMPUS DI CESENA

DIPARTIMENTO DI INFORMATICA – SCIENZA E INGEGNERIA
CORSO DI LAUREA IN INGEGNERIA E SCIENZE INFORMATICHE

Titolo TODO:

TODO

Elaborato in:
TODO

Relatore
TODO

Candidato
TODO

Anno Accademico TODO-TODO

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Abstract

TODO IoT (Internet of Things) è l'acronimo utilizzato per indicare la sfera tecnologica che ha alla base l'utilizzo di oggetti intelligenti. Per oggetti intelligenti si intendono quei dispositivi, macchine o attrezzature che si possono connettere a internet per trasmettere, ricevere dati o elaborarli. All'interno di questa sfera, uno dei metodi di comunicazione più utilizzati è quello basato su un modello Publish/Subscribe che consente lo scambio di dati tra dispositivi della rete senza la necessità di connessioni dirette tra loro. Per funzionare si utilizza il protocollo MQTT, uno dei protocolli più diffusi per lo scambio di messaggi, il quale si basa su una struttura con 3 tipologie di dispositivi connessi: Publisher, Subscriber e Broker. I Publisher una volta ottenuti dei dati tramite sensori di vario genere, li pubblica, rendendoli disponibili nella rete. I dati per spostarsi all'interno della rete utilizzano i broker cioè vari punti/server di connessione, i quali permettono il passaggio di dati per poi infine arrivare ai Subscribers per la lettura e l'elaborazione. Questa tesi riprende il lavoro effettuato da un collega nell'anno 21-22 approfondendo e sviluppando i casi di test con particolare attenzione a quelle che sono le caratteristiche che possono portare criticità al calcolo della soluzione ottima di percorso per il flusso dei dati. La tesi del collega si basa sull'idea di costruire un sistema distribuito sfruttando le proprietà matematiche del rilassamento Lagrangiano, dove ogni nodo conosce solamente i propri vicini.

Chapter 1

Introduction

Write your intro here. _____

You can use acronyms that you defined previously, such as cro:IoTInternet of Thing (IoT). If you use acronyms twice, they will be written in full only once (indeed, you can mention the IoT now without it being fully explained). In some cases, you may need a plural form of the acronym. For instance, that you are discussing cro:vmVirtual Machines (VMs), you may need both VM and VMs.

TODO:
Add side-
notes in
this way.
They are
named af-
ter the au-
thor of the
thesis

Structure of the Thesis

TODO: At the end, describe the structure of the paper

Chapter 2

State of the art

I suggest referencing stuff as follows: fig. 2.1 or Figure 2.1

2.1 Some cool topic

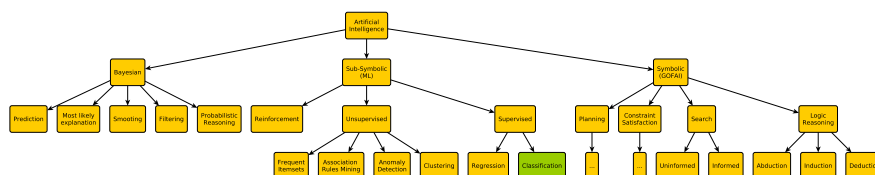


Figure 2.1: Some random image

Chapter 3

Contribution

You may also put some code snippet (which is NOT float by default), eg: chapter 3.

3.1 Fancy formulas here

```
1 public class HelloWorld {  
2     public static void main(String[] args) {  
3         // Prints "Hello, World" to the terminal window.  
4         System.out.println("Hello, World");  
5     }  
6 }
```

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